3.2.2

Wednesday, April 8, 2026 2:41 PM

Exercise 5 (Integer Long Division). Find an $m \in \mathbb{Z}$ and an $r \in \mathbb{N}$ with $0 \leq r < |b|$ such that $a = m \cdot b + r$ holds for the following pairs:

- $(a,b) = (27,5)$
- $(a,b) = (27,-5)$
- $(a,b) = (127,0)$
- $(a,b) = (-1687,11)$
- $(a,b) = (0,7)$

In which cases are your solutions unique?

$(27,5)$ $\quad a = m.b + r$

$27 = 5.5 + 2$

$$\begin{array}{r} 5 \\ 5\overline{)27} \\ 25 \\ \hline 2 \end{array} -$$

$(27,-5)$ $\quad 27 = -5.-5 + 2$

$$\begin{array}{r} -5 \\ -5\overline{)27} \\ 25 \\ \hline 2 \end{array} -$$

$(127,0)$

$b=0$ m,r not exist

Let $a \in \mathbb{Z}$ and $b \in \mathbb{Z}$ be two integers with $b \neq 0$. Then there is always another integer $m \in \mathbb{Z}$ and a non negative integer $r \in \mathbb{N}_0$, with $0 \leq r < |b|$ such that the following holds:

$$a = m \cdot b + r \tag{3.7}$$

$(-1687,11)$ $\quad -1687 = -159.11 + 7$

$$\begin{array}{r} -159 \\ 11\overline{)-1687} \\ -11 \\ \hline -587 \\ -55 \\ \hline -37 \\ -99 \\ \hline 7 \end{array} -$$

$(0,7)$ $\quad 0 = 0.7 + 0$

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